

iClusterBayes Summary Report

Automated Report

Hyperparameters

```
cat('sdev = 0.01, beta.var.scale = 0.3\n')
```

```
## sdev = 0.01, beta.var.scale = 0.3
```

Key Summary Statistics

```
cat(paste0('```\n', '
--- Summary for K = 1 ---
Z.ar:
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  0.6711  0.7056  0.7117  0.7120  0.7194  0.7456

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  0.0000  0.5817  0.7169  0.6622  0.8061  0.9717
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
    1         1         1         1         1         1
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
    1         1         1         1         1         1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
    1         1         1         1         1         1
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
    1         1         1         1         1         1

gamma.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  0.0000  0.5674  0.6983  0.6437  0.7844  0.9472
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  0.01722 0.04500 0.07111 0.23761 0.26722 1.00000
```

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Modality 3 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
Modality 4 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.02556 0.04556 0.06944 0.20808 0.20306 1.00000
Modality 5 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.01556 0.05278 0.13556 0.33924 0.63722 1.00000

--- Summary for K = 2 ---
Z.ar:
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
  0.5361  0.5628  0.5706  0.5708  0.5789  0.6178

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
  0.0000  0.3411  0.5161  0.4767  0.6428  0.9467
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1

gamma.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
  0.0000  0.3344  0.5044  0.4587  0.6183  0.8761
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.00000 0.01167 0.33167 0.48468 1.00000 1.00000
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.000000 0.005556 0.030556 0.314002 0.825833 1.000000
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
  0.0000  0.5056  1.0000  0.7551  1.0000  1.0000

--- Summary for K = 3 ---
Z.ar:
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
  0.3950  0.4233  0.4317  0.4322  0.4406  0.4889

```

```

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.00000 0.05444 0.32389 0.30981 0.49889 0.92667
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1

gamma.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.00000 0.05556 0.32333 0.30010 0.48667 0.81167
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000  0.5189  0.8506  0.7215  1.0000  1.0000
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.8339  1.0000  1.0000  0.9999  1.0000  1.0000
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.000000 0.000000 0.007778 0.258566 0.537639 1.000000
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000  0.5672  0.9756  0.7490  1.0000  1.0000

--- Summary for K = 4 ---
Z.ar:
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.3533  0.3844  0.3933  0.3939  0.4028  0.4489

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.00000 0.07333 0.26722 0.27039 0.42167 0.90889
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 5 :

```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0750	0.2667	0.2596	0.4099	0.7706

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.1494	0.3830	0.8617	1.0000

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.7228	1.0000	1.0000	1.0000	1.0000	1.0000

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.2338	0.4819	1.0000

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.7478	0.9706	0.7898	1.0000	1.0000

--- Summary for K = 5 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.3900	0.4211	0.4317	0.4314	0.4417	0.4878

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.03333	0.18528	0.21042	0.33722	0.89389

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.03556	0.18667	0.20103	0.33000	0.70889

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.1028	0.3158	0.6400	1.0000

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.9461	1.0000	0.9584	1.0000	1.0000

Modality 4 :					
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Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.000000	0.000000	0.118778	0.003472	1.000000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.1839	0.5344	0.5367	0.9728	1.0000

--- Summary for K = 6 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.3250	0.3711	0.3828	0.3848	0.3967	0.4667

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.01944	0.16056	0.19185	0.31167	0.86278

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.02111	0.16333	0.18621	0.31111	0.69333

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.1801	0.3061	1.0000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.2383	1.0000	1.0000	0.9823	1.0000	1.0000

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.00000	0.00000	0.09181	0.00000	1.00000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.1039	0.4028	0.4011	0.6400	1.0000

--- Summary for K = 7 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.3950	0.4411	0.4533	0.4536	0.4650	0.5339

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
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0.00000 0.01125 0.14222 0.18857 0.29667 0.85056

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.01222	0.13778	0.15763	0.26611	0.68667

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.00000	0.00000	0.07117	0.00000	1.00000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.6678	0.8600	0.8137	1.0000	1.0000

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.00000	0.00000	0.0335	0.00000	1.0000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.00000	0.1306	0.2486	0.4383	1.0000

--- Summary for K = 8 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.6183	0.8411	0.8594	0.8549	0.8750	0.9050

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.002778	0.047778	0.131900	0.172778	0.899444

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

```

gamma.ar:
  Modality 1 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.000000 0.003333 0.048333 0.090801 0.150000 0.563889
  Modality 2 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.000    0.000    0.000    0.036    0.000    1.000
  Modality 3 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.0000000 0.0000000 0.0000000 0.0000797 0.0000000 0.5955556
  Modality 4 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.000000 0.000000 0.000000 0.00342 0.000000 1.000000
  Modality 5 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.000000 0.000000 0.000000 0.02731 0.000000 1.000000

--- Summary for K = 9 ---
Z.ar:
  Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.6911 0.8033 0.8211 0.8180 0.8361 0.8756

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.00000 0.00500 0.04778 0.11236 0.14833 0.85778
  Modality 2 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
1      1      1      1      1      1
  Modality 3 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
1      1      1      1      1      1
  Modality 4 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
1      1      1      1      1      1
  Modality 5 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
1      1      1      1      1      1

gamma.ar:
  Modality 1 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.000000 0.005556 0.050000 0.085178 0.136667 0.597222
  Modality 2 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.000000 0.000000 0.000000 0.005525 0.000000 1.000000
  Modality 3 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.00000 0.00000 0.00000 0.13780 0.05778 1.00000
  Modality 4 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.0000000 0.0000000 0.0000000 0.0003795 0.0000000 0.5950000
  Modality 5 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.

```

```
0.0000000 0.0000000 0.0000000 0.0009771 0.0000000 1.0000000
', '\n`'))
```

```
## ' '
##
##
## --- Summary for K = 1 ---
## Z.ar:
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.6711 0.7056 0.7117 0.7120 0.7194 0.7456
##
## beta.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000 0.5817 0.7169 0.6622 0.8061 0.9717
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##    1      1      1      1      1      1
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##    1      1      1      1      1      1
##   Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##    1      1      1      1      1      1
##   Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##    1      1      1      1      1      1
##
## gamma.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000 0.5674 0.6983 0.6437 0.7844 0.9472
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.01722 0.04500 0.07111 0.23761 0.26722 1.00000
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##    1      1      1      1      1      1
##   Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.02556 0.04556 0.06944 0.20808 0.20306 1.00000
##   Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.01556 0.05278 0.13556 0.33924 0.63722 1.00000
##
## --- Summary for K = 2 ---
## Z.ar:
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.5361 0.5628 0.5706 0.5708 0.5789 0.6178
##
## beta.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
```

```

## 0.0000 0.3411 0.5161 0.4767 0.6428 0.9467
## Modality 2 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
## Modality 3 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
## Modality 4 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
## Modality 5 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
##
## gamma.ar:
## Modality 1 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 0.0000 0.3344 0.5044 0.4587 0.6183 0.8761
## Modality 2 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 0.00000 0.01167 0.33167 0.48468 1.00000 1.00000
## Modality 3 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
## Modality 4 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 0.000000 0.005556 0.030556 0.314002 0.825833 1.000000
## Modality 5 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 0.0000 0.5056 1.0000 0.7551 1.0000 1.0000
##
## --- Summary for K = 3 ---
## Z.ar:
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 0.3950 0.4233 0.4317 0.4322 0.4406 0.4889
##
## beta.ar:
## Modality 1 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 0.00000 0.05444 0.32389 0.30981 0.49889 0.92667
## Modality 2 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
## Modality 3 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
## Modality 4 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
## Modality 5 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
##
## gamma.ar:

```

```

## Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.05556 0.32333 0.30010 0.48667 0.81167
## Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000  0.5189  0.8506  0.7215  1.0000  1.0000
## Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.8339  1.0000  1.0000  0.9999  1.0000  1.0000
## Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.000000 0.000000 0.007778 0.258566 0.537639 1.000000
## Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000  0.5672  0.9756  0.7490  1.0000  1.0000
##
## --- Summary for K = 4 ---
## Z.ar:
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.3533  0.3844  0.3933  0.3939  0.4028  0.4489
##
## beta.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.07333 0.26722 0.27039 0.42167 0.90889
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##
## gamma.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000  0.0750  0.2667  0.2596  0.4099  0.7706
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000  0.0000  0.1494  0.3830  0.8617  1.0000
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.7228  1.0000  1.0000  1.0000  1.0000  1.0000
##   Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000  0.0000  0.0000  0.2338  0.4819  1.0000
##   Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000  0.7478  0.9706  0.7898  1.0000  1.0000

```

```

##
## --- Summary for K = 5 ---
## Z.ar:
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.3900  0.4211  0.4317  0.4314  0.4417  0.4878
##
## beta.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.03333 0.18528 0.21042 0.33722 0.89389
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##
## gamma.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.03556 0.18667 0.20103 0.33000 0.70889
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000  0.0000  0.1028  0.3158  0.6400  1.0000
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000  0.9461  1.0000  0.9584  1.0000  1.0000
##   Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.000000 0.000000 0.000000 0.118778 0.003472 1.000000
##   Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000  0.1839  0.5344  0.5367  0.9728  1.0000
##
## --- Summary for K = 6 ---
## Z.ar:
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.3250  0.3711  0.3828  0.3848  0.3967  0.4667
##
## beta.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.01944 0.16056 0.19185 0.31167 0.86278
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.

```

```

##      1      1      1      1      1      1
## Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
## Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##
## gamma.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.02111 0.16333 0.18621 0.31111 0.69333
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000  0.0000  0.0000  0.1801  0.3061  1.0000
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.2383  1.0000  1.0000  0.9823  1.0000  1.0000
##   Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.00000 0.00000 0.09181 0.00000 1.00000
##   Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000  0.1039  0.4028  0.4011  0.6400  1.0000
##
## --- Summary for K = 7 ---
## Z.ar:
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.3950  0.4411  0.4533  0.4536  0.4650  0.5339
##
## beta.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.01125 0.14222 0.18857 0.29667 0.85056
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##
## gamma.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.01222 0.13778 0.15763 0.26611 0.68667
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.00000 0.00000 0.07117 0.00000 1.00000

```

```

## Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000 0.6678 0.8600 0.8137 1.0000 1.0000
## Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000 0.0000 0.0000 0.0335 0.0000 1.0000
## Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000 0.0000 0.1306 0.2486 0.4383 1.0000
##
## --- Summary for K = 8 ---
## Z.ar:
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.6183 0.8411 0.8594 0.8549 0.8750 0.9050
##
## beta.ar:
## Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.000000 0.002778 0.047778 0.131900 0.172778 0.899444
## Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
## Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
## Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
## Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##
## gamma.ar:
## Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.000000 0.003333 0.048333 0.090801 0.150000 0.563889
## Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.000 0.000 0.000 0.036 0.000 1.000
## Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000000 0.0000000 0.0000000 0.0000797 0.0000000 0.5955556
## Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.00000 0.00000 0.00342 0.00000 1.00000
## Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.00000 0.00000 0.02731 0.00000 1.00000
##
## --- Summary for K = 9 ---
## Z.ar:
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.6911 0.8033 0.8211 0.8180 0.8361 0.8756
##

```

```

## beta.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.00500 0.04778 0.11236 0.14833 0.85778
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##
## gamma.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.000000 0.005556 0.050000 0.085178 0.136667 0.597222
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.000000 0.000000 0.000000 0.005525 0.000000 1.000000
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.00000 0.00000 0.13780 0.05778 1.00000
##   Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000000 0.0000000 0.0000000 0.0003795 0.0000000 0.5950000
##   Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000000 0.0000000 0.0000000 0.0009771 0.0000000 1.0000000
##
## ‘‘‘

```

Conclusion

and basic statistics for each K (1–9). The .png files show Z.ar, beta.ar, and gamma.ar distributions.